

A Novel Machine Learning Based Cluster-Then-Forecast Framework for Sensor Placement Optimization and Real-Time Water Level Prediction in Low Impact Development (LID) Stormwater System

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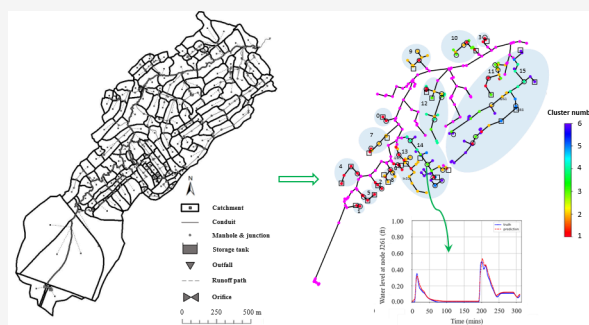
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ABSTRACT: Emerging real-time monitoring technologies have been implemented in stormwater system management in recent decades. However, barriers such as limited monitoring resources and transient reaction time for decision making still exist in practical applications. Nonstationary rainfall and urbanization can relocate the flooding hotspots. To bridge these gaps, a novel machine learning based cluster-then-forecast framework is proposed for sensor placement optimization and real-time water level prediction at both monitored and unmonitored nodes in stormwater systems with Low Impact Development. The developed customized clustering algorithm enables the pinpointing of representative centroid nodes for optimized sensor placement. Past 25 min water level data from monitored nodes is used by LSTM models to predict system-wide water levels with a 5 min lead in the case study, in which centroid node information significantly improves the forecast accuracy for all nodes in a spatially holistic manner. To our knowledge, this study is the first work to predict hydraulic information at unmonitored nodes using only monitored data in real time, addressing the challenge of limited sensor coverage in newly emerging flood hotspots.

KEYWORDS: Stormwater Management, Real-Time Monitoring, Sensor Placement Optimization, Customized Clustering Algorithm, Long Short-Term Memory (LSTM), Low Impact Development (LID), Smart Sewer System



Novel machine learning models to optimize monitoring sensor placement in the stormwater system and to accurately forecast water process at any nodes in the system

1. INTRODUCTION

The urban flooding phenomenon has been aggravated by accelerating urbanization, infrastructural deterioration, and climate change in recent decades. Severe urban floods with large spatial and temporal uncertainties are usually triggered by nonstationary precipitation and low greenery coverage and further exacerbated by sewer malfunctions including sedimentations, cracks, roots, and so on.^{1–4} Traditional urban drainage systems are designed based on statistical records of historical precipitation events.⁵ However, the capability of drainage infrastructure can be weakened continuously as the water infrastructure ages. In this regard, flood hazards such as node inundations and combined sewer overflows are more probable in increasingly unpredictable future climate scenarios, which can lead to significant structural damage, economic loss, and social disruption. Recent studies have explored various flood mitigation strategies, including retrofitted stormwater controls and low-impact development (LID) practices, optimization-based green infrastructure planning, and multistage decision-making frameworks.^{6–8} However, the effectiveness of these approaches depends on accurate system monitoring and adaptive management under changing hydrological conditions. Consequently, obtaining real-time information on the storm-

water system is crucial for developing a control strategy, maintenance planning, and flood risk mitigation.

A real-time monitoring system is a cost-effective stormwater management tool for urban flood mitigation and informed decision-making.^{9–11} The sensing technology can facilitate intelligence acquisition, and then the processed data can be used to direct the operation of controllable components within the stormwater system.¹² More specifically, real-time flow and water level information can provide important guidance for optimizing pump, valve, or storage control strategies and maximizing hydraulic system performance. Real-time monitoring of smart sewer systems typically involves a variety of sensors designed to measure different hydraulic parameters, including water flow, level, and quality. Water level sensors have been widely used in real-time monitoring for overflow detection, which leverages various technologies to ensure

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accurate and reliable measurements.¹³ Ultrasonic level sensors use sound waves to gauge water levels, making them effective for continuous monitoring in manholes or storage tanks.¹⁴ Pressure transducers, on the other hand, measure the water level based on the liquid pressure.¹⁵ Radar sensors employ microwave signals to detect water levels, offering high precision of measurement.¹⁶ Toran used low-cost water level loggers at the inlet and outlet of a small-scale collection system for overflow monitoring.¹⁷ Gaborit used water level sensors to control the outflow of a detention pond located at the outlet of a small urban catchment in Canada, improving removal of pollutants and reducing peak flow dramatically, but this application is limited by the high cost of automatic sensors and data acquisition systems in practice.¹⁸

In terms of large real-world stormwater system monitoring projects, the Metropolitan Sewer District of Greater Cincinnati (MSDGC) has developed a large sewer monitoring program, integrating sensors located at storage areas, treatment facilities, interceptor sewers, and overflow points for combined sewer overflow control. SCADA (Supervisory Control and Data Acquisition), an advanced system that incorporates real-time data collection, monitoring, and control, has empowered MSDGC to significantly cut down overflows from the sewer system by an average of 400 million gallons per year.¹⁹ The City of Los Angeles Bureau of Sanitation has selected sensor locations on major outfalls and interceptors for hydrodynamic model calibration, and Los Angeles County Sanitation Districts installed sensors based on historical information such as past overflow locations.²⁰ Most water utilities reported that monitoring was mainly conducted at the WRRF (Water Resource Recovery Facility) with only a few monitoring sites in the sewer network, and most sensor locations are concentrated on overflow sites, hydraulic facilities, and trunk sewers. Sensor placement strategies in previous academic research also tend to concentrate monitoring resources on more flood-susceptible areas. For example, Li utilized the unsupervised machine learning method to cluster nodes of the drainage network based on similarities of water levels and pinpointed sensors evenly at flood hotspots.²¹ Nonetheless, the spatial and temporal nonstationarity of precipitation in future climate scenarios can induce the relocation of flooding hotspots. Multiple vulnerable sewer nodes near the flood hotspots can become new hotspots under extreme precipitation. Moreover, urbanization is also constantly affecting the potential flooding locations. Therefore, a more adaptable and optimized sensor placement strategy is of great significance for systemic flood mitigation and stormwater system management under climate change and urbanization.

Real-time intelligence, when integrated with hydraulic engineers' expertise, empowers the formulation of optimized system operations and informed decision making. However, the response time between sensing feedback and decision making is limited due to rapid and unpredictable variations in rainfall intensity. These variations can lead to sudden surges in water levels, overwhelming traditional collection systems. The increasing frequency and intensity of precipitation events, driven by climate change, have further exacerbated this challenge, significantly undermining the robustness and reliability of real-time decision making. Typically, it is extremely difficult for traditional hydraulic simulation software to conduct real-time simulation because of huge computational burden and trivial parameter settings.²² Therefore, data-driven methods have been given more attention for making real-time

predictions and extending the response time window for making control strategies. Traditional time series forecasting typically leans on memoryless models, including the autoregressive model. Autoregressive (AR) models use past observations to predict future time series through linear modeling, but they have limited capacity to capture complex, nonlinear patterns and long-term dependencies.²³ Machine learning has gained significant prominence in recent years, excelling at capturing complex nonlinear patterns within big data by learning directly from data structures without relying on predefined rules.²⁴ After training, the model's performance is evaluated on a separate test data set to ensure it possesses decent generalization ability when applied to unseen data sets.

Recurrent Neural Networks (RNNs) are classic machine learning models for sequential data processing and time series forecasting based on their strong ability to capture temporal dependencies, and they have been used as an effective tool for real-time forecasting in the water resources domain.^{25–28} Zhang used Recurrent Neural Network models to predict future water levels of storage tanks, providing more response time for storage control strategies.²⁹ Ishida used LSTM neural networks to predict coastal sea level variations.³⁰ Huang and Lee also adopted LSTM models to predict real-time water levels of nodes in urban drainage systems.³¹ However, most researchers focused on predictions at critical points such as discharge outlets and overflow sites on account of restricted sensing resources.^{32,33} In light of problems of current real-time monitoring and sensor placement strategy mentioned above [1) limited sensing resources; 2) adaptability to nonstationary rainfall; 3) versatility in different land uses and LID practices; 4) expanding time horizon between monitoring and decision making systemically], this study proposes a novel machine learning based framework for sensor placement optimization and real-time water level prediction in Low Impact Development (LID) stormwater systems, which is aimed at capturing systemic water level variations using only limited monitoring resources. A customized K-means algorithm is developed to cluster all nodes in the stormwater system based on historical nodal water profiles and flow accumulation levels to pinpoint the centroid node of each cluster in each monitoring zone of the system. Then, monitored real-time series data can be fed into the Long short-term memory (LSTM) model to make step-ahead water level predictions of all the nodes in the stormwater network, since selected nodes to be monitored are the most representative among all to reflect the hydraulic variations in each cluster. This unsupervised–supervised, data-driven framework is applied to a real stormwater system in Cincinnati, Ohio, USA, under nonstationary rainfalls and distinct Low Impact Development (LID) scenarios.

2. METHODOLOGY

2.1. Cluster-then-Forecast Framework. This study proposes a novel, data-driven, and customized cluster-then-forecast framework for sensor placement optimization and real-time water level prediction with limited monitoring resources. As shown in Figure 1, this framework includes two main parts: an optimized sensor placement strategy and real-time forecasting based on hydraulic simulations. Identifying critical and significant nodes for prior sensor localization is the primary step of the strategy, which is based on comprehensive analysis of nodes' significance in the stormwater system testbed (Storm Water Management Model) considering flood susceptibility, maintenance level, and functional significance.

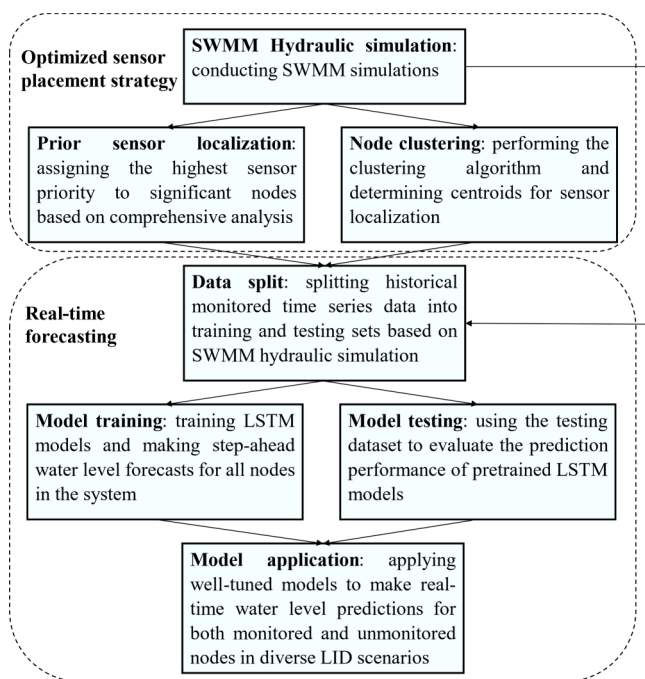


Figure 1. General procedures of the cluster-then-forecast framework.

Flood susceptibility is evaluated through an analysis of historical flooding incidents, with a focus on identifying flooding hotspots. This assessment incorporates use of historical rainfall data to pinpoint nodes that are vulnerable to flooding. High-risk nodes that have experienced frequent or severe flooding in the past are given a higher priority for sensor placement to enable effective real-time monitoring for potential flood detection and timely maintenance schedules. Other factors concerning maintenance history, inspection records, cleaning schedules, specific functions such as outfall, Combined Sewer Overflow (CSO) point, or major interceptors are also considered as significant indicators in prioritizing sensor placement, since monitoring these nodes can help prevent system failures and ensure the continuous operation of essential hydraulic functions. After localizing prior sensor locations, the testbed model in SWMM5 is simulated by using historical rainfall data to extract time series water level data of nodes as the input into the customized K-means

clustering algorithm. During the clustering stage, the whole system is first partitioned into different monitoring zones based on physical proximity and system topology. Then, all nodes of each monitoring zone are classified into several clusters according to their intrinsic characteristics such as node order and historical nodal water profiles. By clustering, all nodes in each cluster possess similar intrinsic characteristics and water level patterns. Centroid information on each cluster can guide the localization of additional sensors. More specifically, the node that is the closest to the centroid point of each cluster in terms of Euclidean distance in feature space is chosen as the monitoring location. In this regard, all centroid information can be integrated to reflect the water level variations of the whole system. Centroid nodes identified by the clustering algorithm are the second level prior locations for sensor placement. The optimized sensor placement strategy broadens the monitoring coverage to contain not only flooding hotspots but also key nodes that reflect overall water level fluctuations throughout the system, based on which the real-time forecasting model can be constructed to predict water profiles of not only overflow-susceptible nodes but also unmonitored nodes. The second part of the framework is about real-time forecasting based on hydraulic simulations. Initially, the input matrix including the monitored water level series, rainfall series, and node features is generated by hydrological and hydraulic simulation in SWMM5. Based on the generated data matrix, the Long short-term memory (LSTM) model is trained to have sufficient generalization ability for making systemic, step-ahead, real-time water level predictions in testing rainfall series with limited monitored information. Finally, pretrained LSTM models are implemented to make real-time water level forecasts of both monitored and unmonitored nodes in new real-world rainfall events under different LID-control scenarios.

2.2. Customized K-Means Algorithm. Unsupervised learning algorithms have been widely used in water distribution networks for leak detection; however these algorithms are rarely used in studies of stormwater system management.^{34–36} Sun used the K-means algorithm to optimize the input parameters into the SWMM model.³⁷ Li adopted three clustering methods to discover the dissimilarities of different water depth series data, but the impact of rainfall uncertainty needs further exploration.³⁸ However, nonstationary precipitation and urbanization can relocate overflow hotspots.

Table 1. Pseudocode of Customized K-Means Clustering Algorithm

Algorithm: Customized K-means clustering algorithm for stormwater system partition	
Input: the feature matrix: $n \times m$ (n is the number of data points); the number of clusters: k ; the number of features: m ; the number of maximum iterations: $\max_iter$; the weight of intrinsic sewer nodal characteristics: ω ; the weight of nodal water level profiles: $1 - \omega$	
Output: clusters and centroid points	
Step 1: Initialize input parameters.	
Step 2: Assign random values to the centroid matrix: $k \times m$; k centroid feature vectors are corresponding to k clusters.	
Step 3: Calculate the Euclidean distance between each data point and k centroid feature vectors, respectively.	
Step 4: Assign each data point to one of k clusters based on which centroid feature vector is the closest to this data point in the Euclidean feature space. The Euclidean distance is calculated as	
$D_{i,j} = \omega d_{1,i,j} + (1 - \omega) d_{2,i,j}$	
where $D_{i,j}$ is the total Euclidean distance between i th data point and j th centroid feature vector; $d_{1,i,j}$ is the nodal order distance between them computed based on the first two features: Nodal Strahler order and Shreve order; $d_{2,i,j}$ is the Euclidean distance between them computed based on features of nodal water level profiles: nodal water level series at a certain period of time, ω is the weighting factor.	
Step 5: Update the centroid matrix $k \times m$ by averaging the feature vectors of data points in each cluster.	
Step 6: While $\max_iter > 0$, repeat steps 3 and 4, and subtract one from $\max_iter$ every time; proceed with the iteration unless one of the two requirements is met: 1) $\max_iter$ is not larger than zero; 2) the centroid matrix remains unchanged.	
Step 7: Return clustering results and the centroid point of each cluster.	

Considering these limitations of previous studies, an innovative customized K-means clustering algorithm was developed to group nodes of a stormwater network based on nodal flow accumulation levels and historical water level time series, guiding optimized sensor localization strategies when facing changing climate and urbanization. A novel function of the proposed algorithm is the capability of flexible weight adjustments to a single input feature or multiple features as a whole to represent a physical meaning. The closest sewer node to the centroid point of each cluster in terms of Euclidean distance in the feature space is assigned as the additional monitoring location, apart from the prior sensor locations. The monitored data at these selected nodes are used to predict water level variations at other nodes of corresponding clusters, since each cluster has strong feature similarities.

The K-means algorithm can partition all data points into k clusters, in which each cluster has small feature variance.³⁹ It is used to characterize highly susceptible flooded areas in traditional studies; however, the relocation of flooding hotspots because of uncertain climate or urbanization constrains the applications of these studies. Thus, this study proposes an adaptable K-means algorithm that enables adjustments to the weights of both intrinsic characteristics of the stormwater network and dynamic water patterns of nodes: nodal order and water level profiles at a certain period of time. After the algorithm is performed, the sewer nodes within each cluster have similar features concerning flow accumulation levels and water level patterns. Then, the closest node to the centroid point of this cluster in feature space is selected as the second level prior to location for sensor installation. In this regard, the monitored information can reflect the water level variations of other nodes in this cluster to the maximum extent, despite the flood hotspot relocation due to climate change or urbanization.

The specific procedures of the customized K-means clustering algorithm are illustrated in Table 1. The inputs and outputs of this algorithm are stated at the beginning of the pseudocode. After initializing the input parameters, k centroid feature vectors corresponding to k clusters are assigned random values in the feature space, creating a $k \times m$ centroid matrix. Then, each nodal data point is assigned to the corresponding cluster based on the Euclidean distance between this data point's feature vector and the centroid feature vector of this cluster. The first two features of the vector are nodal Strahler order and Shreve order, while the remaining features are water level series at each time step. When computing the total Euclidean distance after normalization, the weight of the nodal order and water level series can be adjusted to uncover the impact of each factor on classification and prediction results. Afterward, the centroid matrix is recalculated by averaging the feature vectors of nodal data points in each cluster. The reassignment and recalculation are iterated until the maximum iteration number is reached, or the centroid matrix remains unchanged. Eventually, the nodal data points of each cluster are returned to localize k nodes that are the closest to k centroid feature vectors in the Euclidean feature space to be monitored for real-time water level forecasts.

3. RESULTS

3.1. Clustering Results. Based on the first step of optimized sensor placement strategy, an initial set of sensors has been assumed to be installed priorly at the flood susceptible nodes that were identified by SWMM5 hydraulic

simulations using historical rainfall series with return periods of 5, 25, and 50 years. These overflow-susceptible nodes are denoted by black squares in Figure 2 and are characterized by

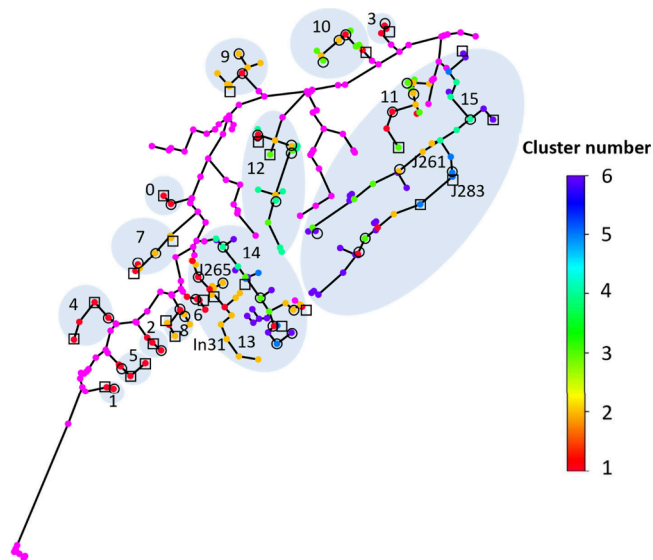


Figure 2. Clustering results of the stormwater system. Black squares denote overflow manholes; black circles denote centroid manholes; blue ellipses denote monitoring zones. Four denoted manholes, J265, In31, J283, and J261, are selected for specific discussion in this work.

historical nodal overflow causing at least 0.2 m of surface inundation. After pinpointing the prior water level sensor locations, a customized K-means clustering algorithm was performed for the second level prior sensor localization. As the initial step of partition, 16 monitoring zones denoted by blue ellipses were selected to conduct the clustering analysis respectively based on physical proximity and system topology as shown in Figure 2. Junctions of open channels denoted by pink were not considered to be clustered in this study. Sixteen groups of sewers that converge into the main open channel were numbered from 0 to 15 as different monitoring zones. The clustering algorithm was executed in each monitoring zone for node clustering. Centroid nodes identified by the algorithm are denoted by black circles, representing corresponding clusters with the same colors. In each monitoring zone, sewer nodes were clustered into several groups represented by different colors, as shown in the color bar. Several monitoring zones from number 0 to 6 possessing a small quantity of junctions were characterized as a single cluster, while monitoring zone 16 with the highest number of nodes was partitioned into 6 clusters. The number of clusters defined in the clustering algorithm for each monitoring zone depended on the amount of limited investable monitoring resources, and it can be tailored to meet customized requirements. Clustering distribution demonstrates that nodes in trunk sewers tend to be clustered in the same group. Nodes in tributary sewers that are geographically close have a tendency to be classified as one cluster. The clustering results verify that sewer nodes that have similar network-wise characteristics and water level patterns can be grouped, and the overall hydraulic condition of each cluster can be captured and represented by the centroid node.

3.2. Prediction Results. Figures 3 and 4 illustrate simulated truth and predicted water level series of four

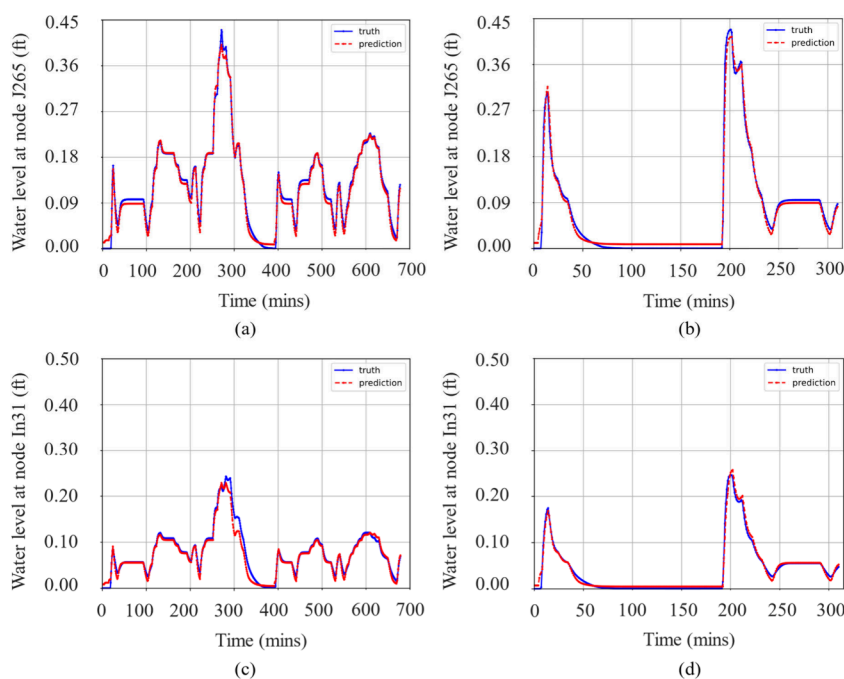


Figure 3. Simulated truth and predicted water level time series of both monitored (J265) and unmonitored (In31) nodes for training data set (a, c) and testing data set (b, d).

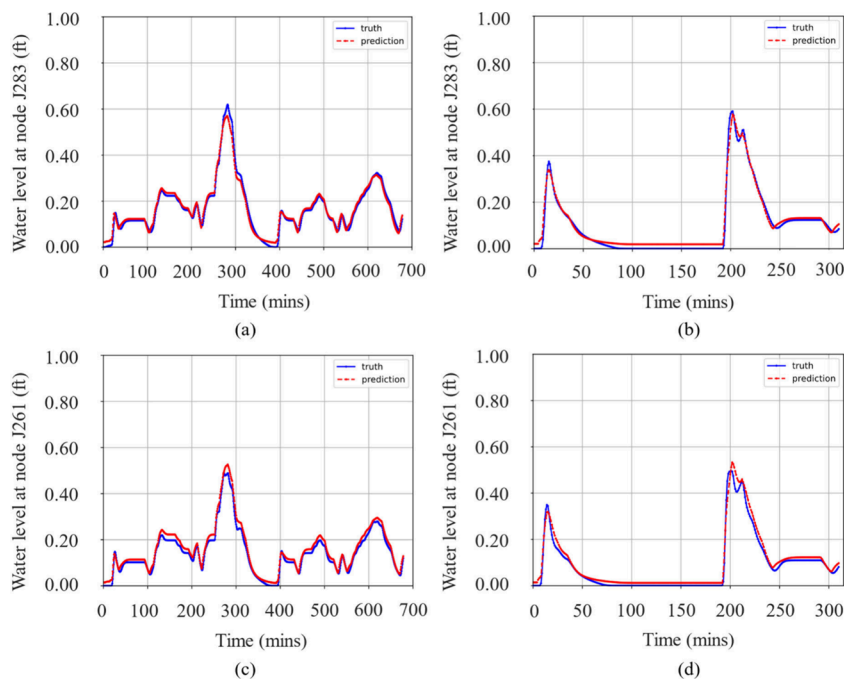


Figure 4. Simulated truth and predicted water level time series of both monitored (J283) and unmonitored (J261) nodes for training data set (a, c) and testing data set (b, d).

selected nodes on training and testing rainfall data sets. Manhole J265 is located at the downstream trunk sewer of monitoring zone 13, which was identified as the centroid node through the customized K-means clustering algorithm. The predicted water profiles of node J265 capture the fluctuations of ground truth with high accuracy ($R^2 = 98.94\%$; RMSE = 0.0096 feet) on the training data set and strong fit ($R^2 = 98.74\%$; RMSE = 0.0127 feet) on the testing data set. For node In31 that is unmonitored and predicted based on monitored information by the LSTM model in the same monitoring zone,

the LSTM neural network was also capable of forecasting the abrupt variations of water levels with decent accuracy under both training ($R^2 = 96.13\%$; RMSE = 0.0098 feet) and testing ($R^2 = 98.44\%$; RMSE = 0.0071 feet) precipitation conditions. Figure 4 shows that the LSTM model can accurately predict water level profiles for both monitored and unmonitored nodes in zone 15 as well, which contains the highest number of nodes, 5 min ahead in real time. For monitoring zone 13, water level series of three nodes (one overflow and two centroid nodes) were inputted into the LSTM neural network to predict

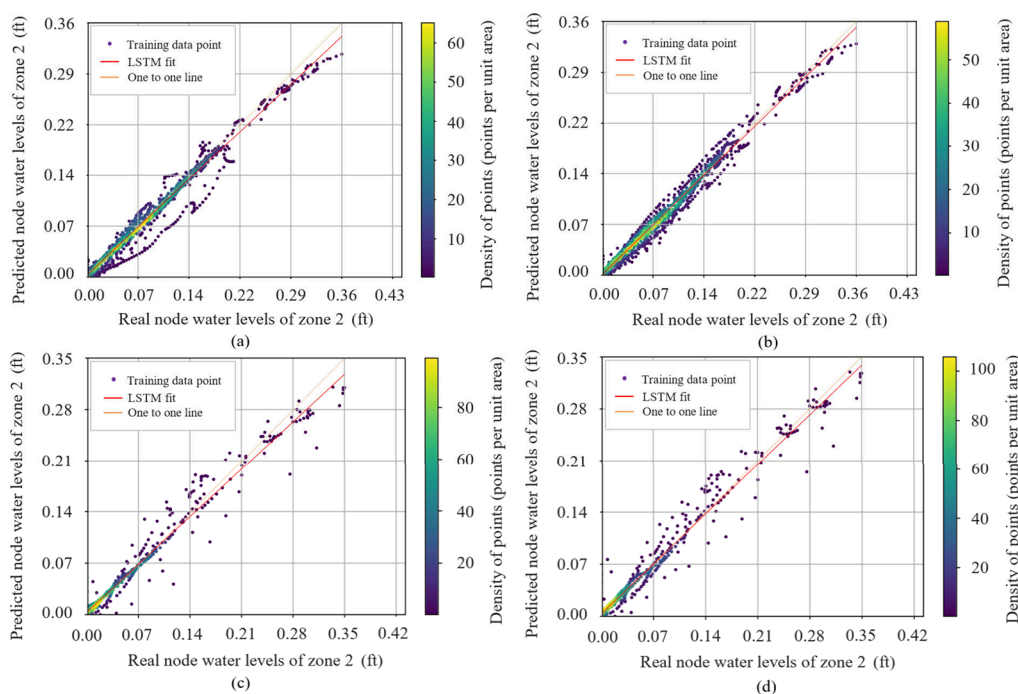


Figure 5. Simulated truths versus predicted water levels of all nodes in specific zones: (a) training data set of zone 2 predicted by overflow and random nodes; (b) training data set of zone 2 predicted by overflow and centroid nodes; (c) testing data set of zone 2 predicted by overflow and random nodes; (d) testing data set of zone 2 predicted by overflow and centroid nodes.

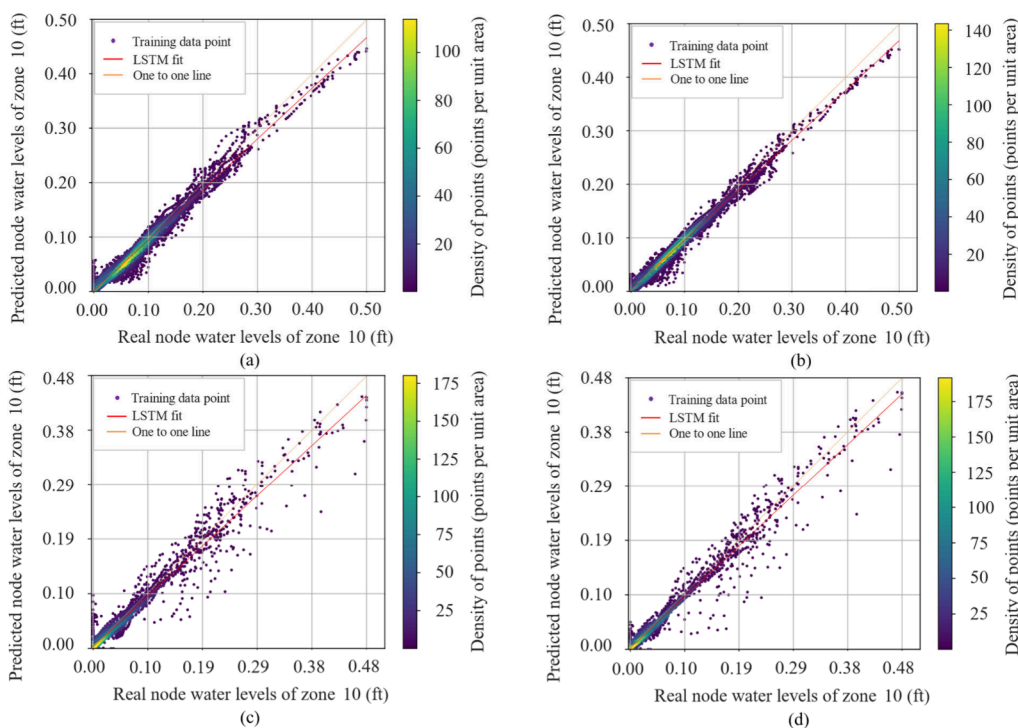


Figure 6. Simulated truths versus predicted water levels of all nodes in specific zones: (a) training data set of zone 10 predicted by overflow and random nodes; (b) training data set of zone 10 predicted by overflow and centroid nodes; (c) testing data set of zone 10 predicted by overflow and random nodes; (d) testing data set of zone 10 predicted by overflow and centroid nodes.

the total 17 nodes' future water depths in this zone, while water level variations of ten nodes (4 overflow and 6 centroid nodes) were used to make step-ahead forecasts on the total 42 nodes in monitoring zone 15, in which the evaluation metrics also substantiate robust prediction performance of LSTM models on selected flood susceptible node J283 and

unmonitored node J261 without overfitting problems. Minor deviations are observed between predicted peak water level values and ground truths, which are negligible in terms of qualitative assessment of real-time flood warnings. Remarkable forecasting performance using new testing rainfall series demonstrates the strong generalization ability of the pretrained

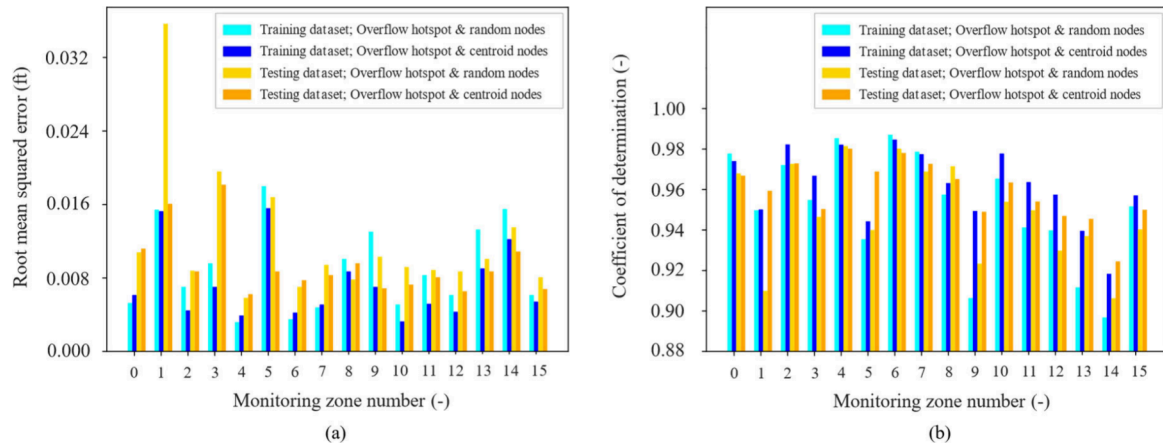


Figure 7. Prediction results in the stormwater system of Shayler Crossing Watershed: (a) mean squared error; (b) coefficient of determination of all monitoring zones.

Table 2. Prediction Results of Zones 10 and 12 under LID and No LID Scenarios^a

Scenario 1 (LID)	Monitoring zone 10				Monitoring zone 12			
	Train I	Train II	Test I	Test II	Train I	Train II	Test I	Test II
R^2	0.9754	0.9778	0.9540	0.9635	0.9599	0.9675	0.9398	0.9550
RMSE	0.0126	0.0101	0.0149	0.0141	0.0118	0.0107	0.0135	0.0116
Scenario 2 (no LID)	Monitoring zone 10				Monitoring zone 12			
	Train I	Train II	Test I	Test II	Train I	Train II	Test I	Test II
R^2	0.9740	0.9806	0.9594	0.9639	0.9548	0.9683	0.9415	0.9528
RMSE	0.0115	0.0098	0.0151	0.0142	0.0124	0.0104	0.0138	0.0124

^aTrain I denotes using overflow node profiles and training rainfall data as inputs; Train II denotes using overflow and centroid node profiles and training rainfall data as inputs; Test I denotes using overflow node profiles and testing rainfall data as inputs; Test II denotes using overflow and centroid node profiles and testing rainfall data as inputs.

LSTM model of each monitoring zone. The cluster-then-forecast framework was proven to uncover water level variations at unmonitored nodes accurately for making optimized hydraulic controls and flood mitigation strategies with limited sensing resources.

Figures 5 and 6 illustrate the correlation between simulated truths and predicted node water levels by the LSTM model on both training and testing data set of all sewer nodes in monitoring zones 2 and 10, respectively. A significant improvement of real-time forecasts on the training rainfall data set is observed in Figure 5a,b. By incorporating the centroid node information, the prediction error (RMSE) was reduced from 0.0150 to 0.0112 ft and R^2 was improved from 97% to 98%. It is clear that there are more data points concentrated along the one-to-one line in Figure 5b than those in Figure 5a, while this accuracy improvement is minor on the testing data set. Similar improvements are also observed in the prediction outcomes of monitoring zone 10 as shown in Figure 6a–d. It can be seen obviously that the water level data points were more concentrated and less scattered along the one-to-one linear line in centroid-node scenarios compared with the baseline scenario, especially when water level values were quite high. It reveals that using centroid water level profiles as model inputs can enhance the forecast accuracy significantly when predicting unmonitored real-time water level variations under the constraint of limited sensing resources.

Prediction results of all monitoring zones are summarized in Figure 7. Two indexes, root mean squared error and coefficient of determination, were used to evaluate the performance of LSTM model predictions. It is obvious that providing centroid

node data for the model can improve the accuracy of real-time forecasts dramatically for monitoring zones with a relatively large number of nodes from 8 to 15, whereas the enhancement is not significant for monitoring zones 0 to 7 with small amounts of nodes. This is because using water level time series at only overflow hotspots as LSTM inputs is capable of predicting the water level variations in the whole zone with decent accuracy: overall R^2 over 90% when the monitoring zone was characterized with a simple network structure and a small number of nodes, which indicates that it is not necessary to utilize additional monitoring resources as extra inputs for the LSTM model to make real-time predictions for the entire zone. On the contrary, a significant improvement can be seen by inputting additional centroid information into the model when predicting the zones with complex topology and considerable nodes. The average root mean squared error reductions for training and testing data sets are 0.020 feet and 0.014 feet (reduction ratio: 28.45%; 14.30%), while R^2 improvements are 1.95% and 1.08%, respectively. It is evident that investing additional monitoring resources at centroid nodes localized by a customized clustering algorithm can help reveal the unmonitored water level profiles. Accordingly, the proposed cluster algorithm and LSTM neural network can effectively enable real-time monitoring and forecasting with a global perspective on spatial coverage. Table 2 presents the real-time forecasting results of all nodes in monitoring zones 10 and 12 under LID and non-LID scenarios. In scenario 1, vegetative swales with average initial saturation 25% are applied to alleviate the adverse impact of peak runoff in most catchments and the drain flow is set to return to previous areas,

Table 3. Sensitivity Analysis of Classification Weight on Prediction Results of the Four Monitoring Zones

	Weight of node order number: ω					
	RMSE ($\omega = 0.4$)		RMSE ($\omega = 0.5$)		RMSE ($\omega = 0.6$)	
	Training	Testing	Training	Testing	Training	Testing
Monitoring zone 12	0.0116	0.0142	0.0116	0.0143	0.0117	0.0141
Monitoring zone 13	0.0160	0.0157	0.0152	0.0158	0.0160	0.0157
Monitoring zone 14	0.0194	0.0183	0.0198	0.0184	0.0197	0.0186
Monitoring zone 15	0.0135	0.0154	0.0135	0.0154	0.0124	0.0150

while no measures of Low Impact Development are used in scenario 2. The area of vegetative swales of each subcatchment ranges from 12.4 to 5915.8 square feet throughout the stormwater system and the input parameter into LSTM model, impervious catchment area directed to each node, was adjusted based on the percentage of impervious area treated by green infrastructure. The results indicate that the LSTM model exhibits excellent prediction performance in both scenarios, proving the adaptability of the proposed cluster-then-forecast model under diverse impervious area circumstances. Incorporating water level information from centroid nodes significantly decreases prediction errors in both scenarios, which demonstrates the effectiveness of the optimized sensor placement strategy.

4. DISCUSSION

The customized K-means clustering algorithm can be adaptive to adjust classification parameters to fulfill practitioners' demands. Specifically, the algorithm is adaptable for decision makers to define the cluster quantity required to install the corresponding water level sensors at the centroid nodes of each cluster. Theoretically, more defined clusters can lead to more accurate prediction results because more information is input into the LSTM model. Nevertheless, monitoring resources are usually constrained due to budget limitations. Installing sensors at all sewer nodes is unrealistic. Thus, the proposed cluster-then-forecast framework is designed to facilitate capturing water level variations at all nodes, especially potential overflow nodes, while conserving sensing resources. Table 3 presents the sensitivity analysis results of the weight impact on the LSTM predictions. The weight of the sewer node order number is adjusted to 0.4, 0.5, or 0.6, respectively. Accordingly, the weight of historical water level profiles is set as 0.6, 0.5, or 0.4 in the customized K-means algorithm. The relative root mean squared error between ground truths and predicted 5 min ahead water levels is far smaller than 1% without overfitting issues in the four monitoring zones on both training and testing rainfall data sets. Based on the comparison of three different weight scenarios, the influence of weight changes on clustering and forecasting results is not significant, which proves that sewer node orders and historical water level profiles play equally important roles in identifying representative centroid nodes for additional sensor placement. This result of sensitivity analysis suggests the intrinsic physical correlation between the two clustering indicators: nodes along trunk sewers tend to exhibit more homogeneous water level fluctuations, while nodes within tributary sewers also share relatively consistent patterns. However, the water level variations between trunk and tributary sewer nodes differ significantly. Consequently, whether clustering is primarily driven by sewer node order or historical water level profiles, the algorithm consistently differentiates trunk sewer nodes

from tributary sewer nodes. Customized feature weighting of the developed clustering algorithm can shed light on informative feature selection if more intrinsic sewer characteristics are incorporated. The developed clustering algorithm enhances interpretability by allowing explicit adjustments of the relative importance of different clustering indicators. By tuning the weights, the algorithm provides transparency in how these factors influence clustering results, enabling a clearer understanding of their hydrodynamic roles.

5. CONCLUSION

A novel machine learning based cluster-then-forecast framework is proposed in this study for sensor placement optimization and real-time water level prediction in stormwater systems. The framework was applied to a real-world case study: Shayler Crossing watershed, Cincinnati, Ohio, USA. The conclusions are made as follows based on the results:

- 1) The optimized sensor placement strategy can guide the optimal sequence of sensor localization, in which historical flood susceptible nodes are pinpointed to be priorly monitored. The customized K-means clustering algorithm is developed to identify the centroid node of each cluster based on the similarities of nodal water level profiles and flow accumulation levels as the second level priority of sensor installation, which aims at localizing the most representative nodes that can reflect the overall water level variation of the whole monitoring zone. The clustering result of the SHC stormwater system shows that nodes at trunk sewers tend to be classified as the same cluster characterized with high order values and similar historical water level patterns, while clusters at tributary sewers are spatially distributed in a scattered way.
- 2) To the best of our knowledge, this is the first work to predict unmonitored node hydraulic information based on monitored data to address the issue of insufficient monitoring coverage of new potential flood hotspots due to limited sensing resources. The LSTM model is used to make real-time water level forecasts with a 5 min lead of both monitored and unmonitored nodes based on only monitored water level and rainfall series data of the past 25 min, and intrinsic characteristics of predicted nodes. Incorporating centroid node data into the LSTM model prediction reduces the average error and improves the coefficient of determination significantly when making real-time water level forecasts on both monitored and unmonitored nodes in a spatially holistic manner, which demonstrates the effectiveness of the cluster-then-forecast framework.
- 3) The prediction accuracy on both training and testing real-world, nonstationary rainfall data sets of monitored nodes was extremely high (R^2 over 97%), while the

accuracy of unmonitored nodes was decent as well (R^2 over 90%), especially for nodes in small monitoring zones with uncomplicated structures. Excellent accuracy was also achieved across various impervious percentage catchments in both LID and non-LID scenarios, which proves that this framework can be adaptable to different land use and LID practice cases. Forecasting water levels at both monitored and unmonitored nodes with limited sensors in a timely and accurate manner enables economic and proactive management of stormwater infrastructure, allowing operators to make prompt and advance controls to mitigate overflow risks. The proposed sensor placement strategy can guide the optimal sequence of sensor localization for hydraulic performance monitoring, rational allocation of sensing resources, and capital improvements. The forecasting model enhances real-time data analytics and quantitatively reveals future water profiles at unmonitored nodes, thereby improving the resilience and intelligence of stormwater management strategies. Integrating this framework with adaptive stormwater management policies, such as watershed-scale flood mitigation strategies and green–gray infrastructure planning, could further expand its applicability to larger scale or nonurban–urban watersheds.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsestwater.4c01277>.

Node ordering, LSTM model structure, case study and hydraulic simulations, clustering processing, data processing and model structure, sensitivity analysis, and prediction results (PDF)

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Author Contributions

CRedit: Enpei Chen data curation, investigation, visualization, writing - original draft.

Notes

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